Negative indices

1

a i 10 ii -7

b i 2 ii -4

c i 13 ii -10

d i -5 ii -8

2

a

2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}
32	16	8	4	2	1	$\frac{1}{2}$

b

10^5	10^4	10^3	10^2	10^1	10^0	10^{-1}
$100\,000$	$10\,000$	1000	100	10	1	$\frac{1}{10}$

c

3^3	3^2	3^1	3^0	3^{-1}	3^{-2}	3^{-3}
27	9	3	1	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{1}{27}$

3

a $\frac{1}{6^{10}}$ **b** $\frac{1}{73^{14}}$ **c** $\frac{1}{(-9)^7}$ **d** $\frac{1}{9}$

e $\frac{1}{(17)^6}$ **f** $\frac{1}{55}$ **g** $\frac{1}{(-12)^8}$ **h** $-\frac{1}{45^5}$

i $-\frac{1}{8^{11}}$ **j** $\frac{1}{(-20)^3}$ **k** $\frac{1}{7^6}$ **l** $\frac{1}{-5}$

4

a 3^{-1} **b** 37^{-1} **c** 5^{-1} **d** 4^{-7}

e -15^{-3} **f** 10^{-5} **g** $(-24)^{-10}$ **h** 25^{-3}

i 13^{-11} **j** 7^{-8} **k** 16^{-12} **l** $(-45)^{-7}$

5

a -2 **b** -5 **c** -3 **d** -3

e -11 **f** -3 **g** -4 **h** -5

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6

a 5^{14} **b** 7^{-12} **c** 7^{-7} **d** 5^{5}

e 9^{-12} **f** 5^{22} **g** 10^{-1} **h** 100^{-1}

i 2^{-2} **j** 5^{-16} **k** 19^{18} **l** -8^{24}

7

a $\frac{13}{9}$ **b** 16 **c** $\frac{19}{18}$ **d** $\frac{81}{4}$

8

a 0

b $0^{-4} = \frac{1}{0^4} = \frac{1}{0}$ and dividing by zero is impossible.

9

a 1 **b** 16 **c** 27 **d** $\frac{1}{25}$

e $\frac{1}{4096}$ **f** $\frac{1}{144}$ **g** $\frac{1}{225}$ **h** $\frac{1}{256}$

i $\frac{1}{49}$ **j** $\frac{1}{9}$ **k** 125 **l** $\frac{1}{-7}$

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